

ARTIFICIAL INTELLIGENCE

Presented By

Dr. KSHITIJ TRIPATHI

Assistant Professor

B.Sc. (Maths), M.Sc, MCA, C-DAC

PhD(Data Science+Artificial Intelligence+Machine Learning)

UGC SLET (COMPUTER SCIENCE)

LOGIC & KNOWLEDGE BY MACHINE

VISION
SPEECH
HEAR

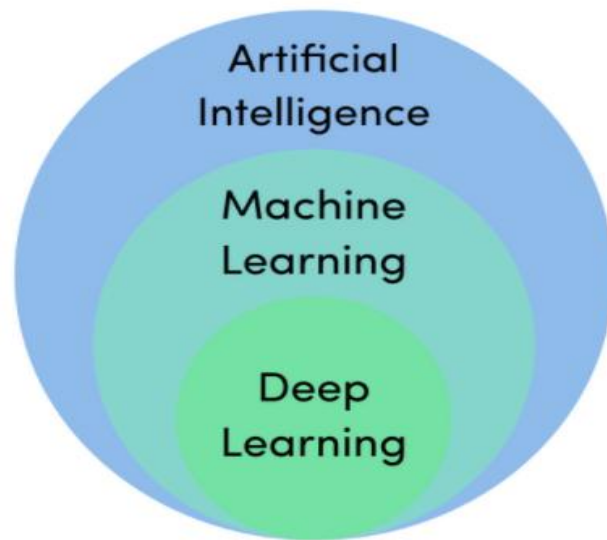
PROCESS CPU -IT HAS BRAIN

PROGRAM

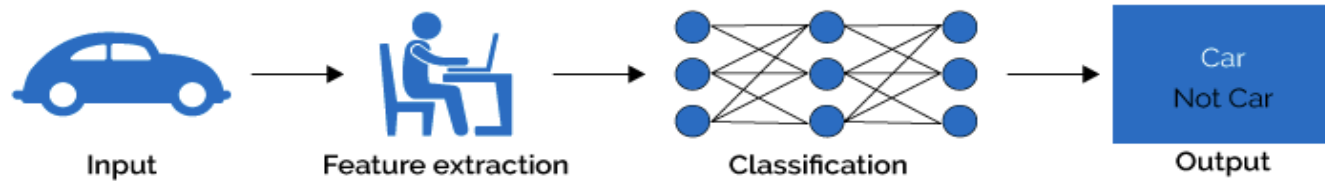
WITHOUT DATA PROGRAM HAS NO MEANING

IN SUMMARY IT IS ALL ABOUT DATA AND ALGORITHMS

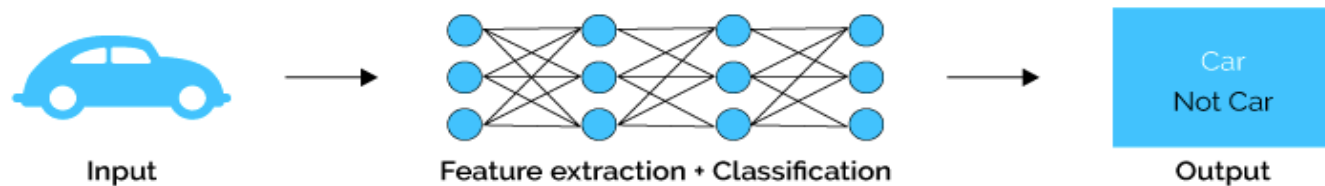
KEY ROLE IS PLAYED BY MATHEMATICS

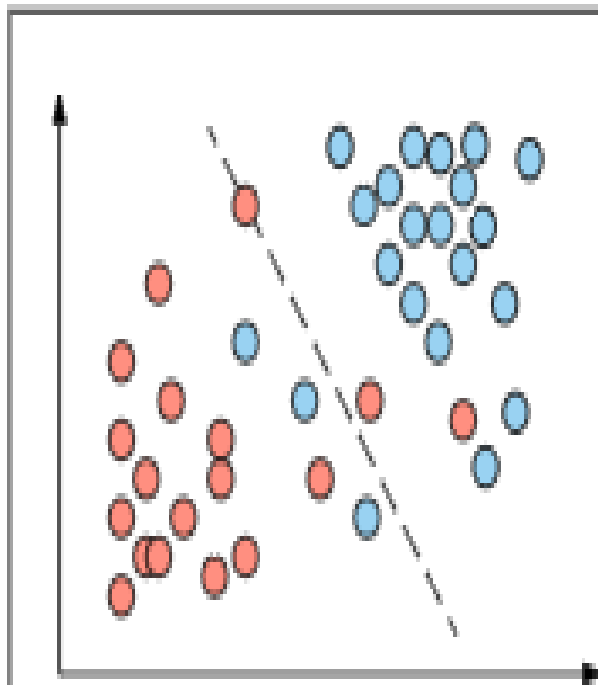


Machine Learning

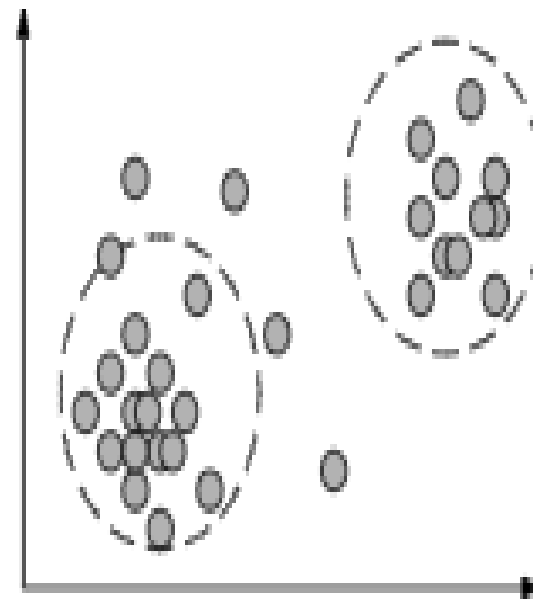


Deep Learning



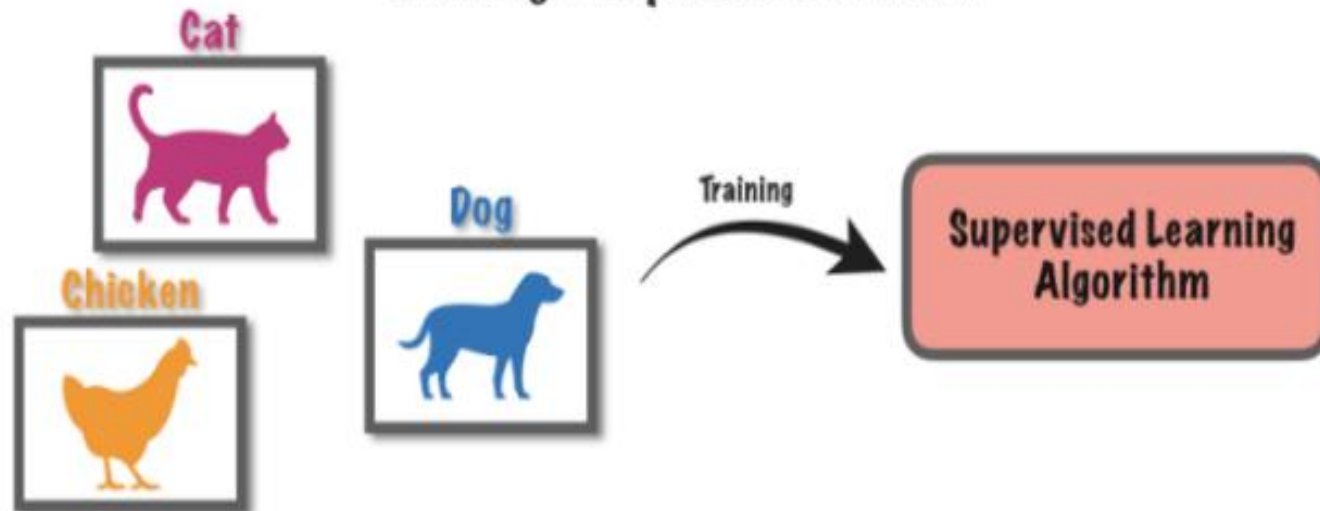


Supervised learning

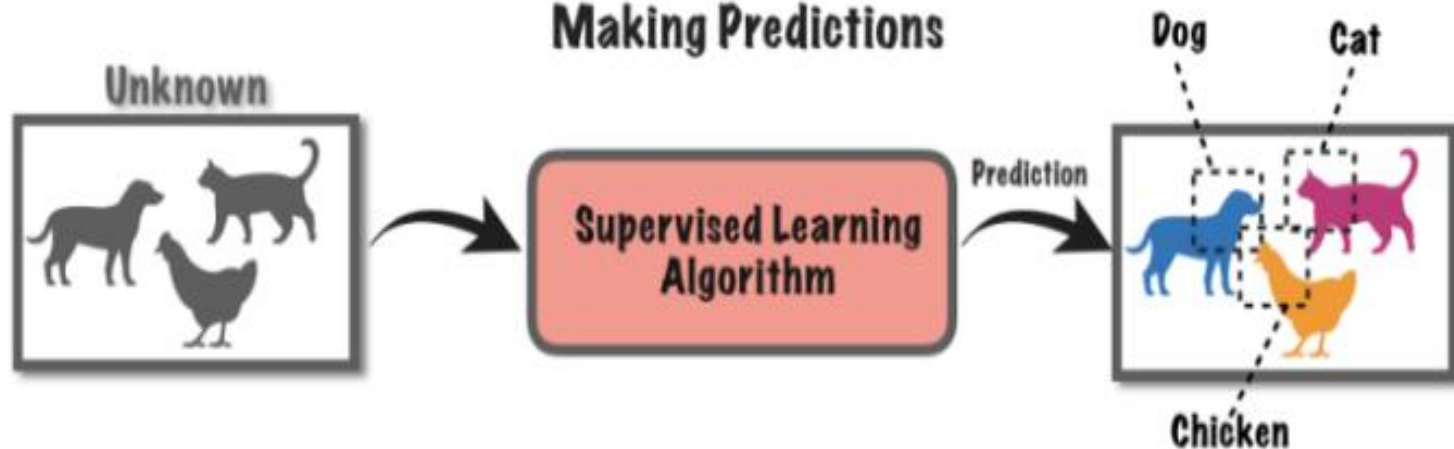


Unsupervised learning

Training a Supervised Learner

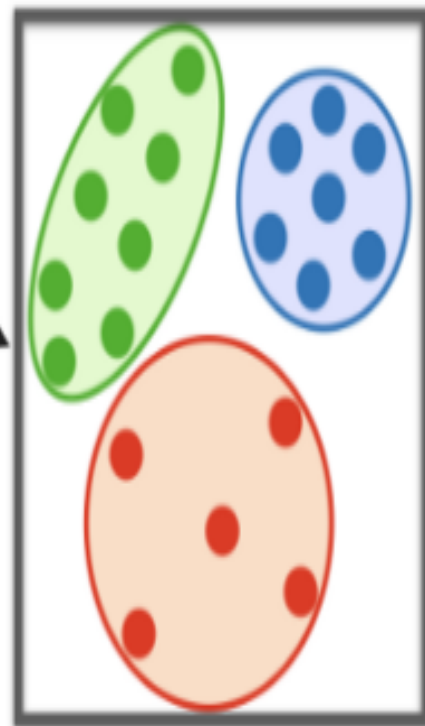


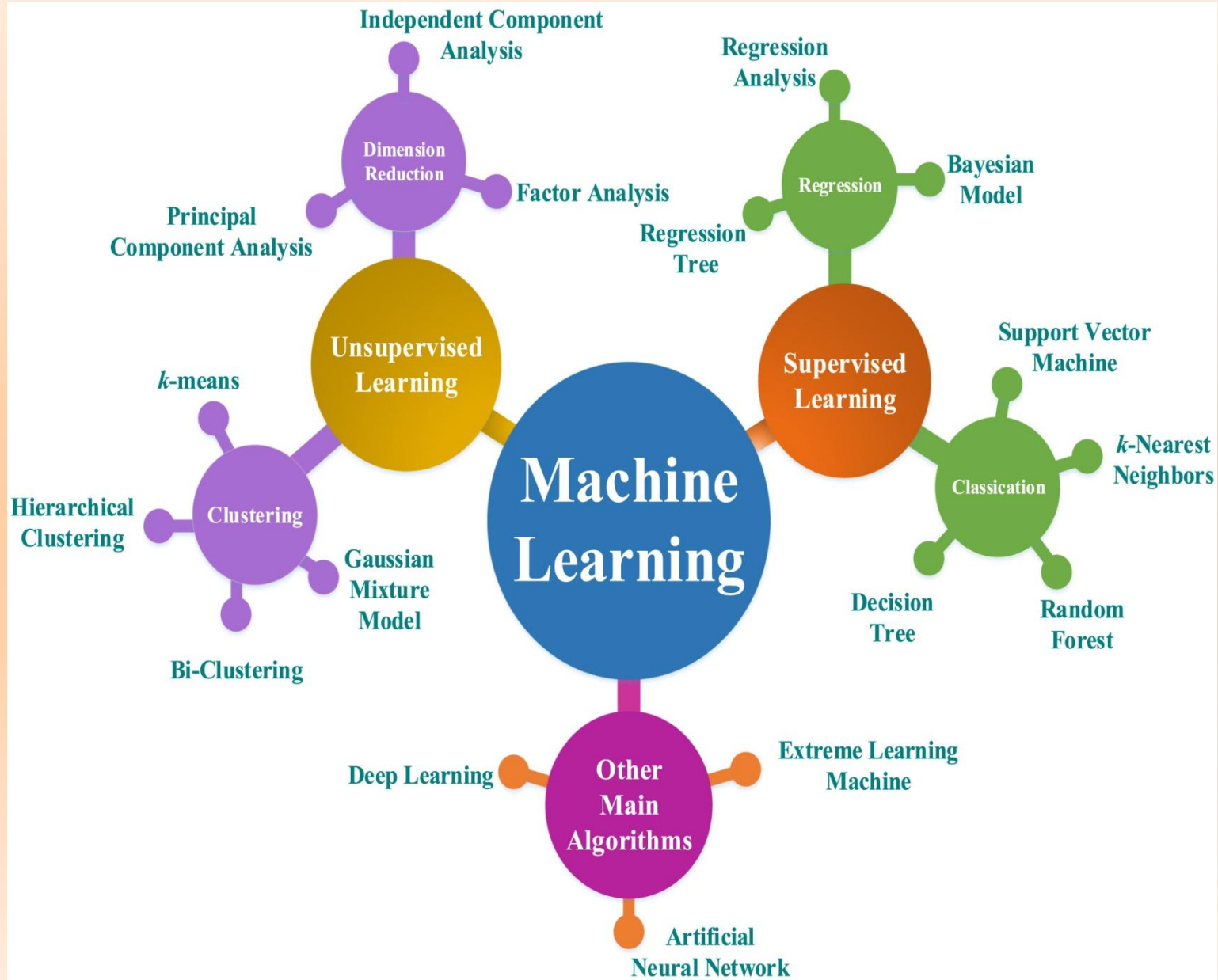
Making Predictions





**Unsupervised Learning
Algorithm**





ADVANTAGES OF MACHINE LEARNING

- **WHOLE PROCESS IS AUTOMATED**
- **PROCESS SMALL AND LARGE DATA ADEQUATELY**
- **HANDLES A VARIETY OF DATA**
- **IDENTIFY THE DATA PATTERNS (SUPERVISED AND UNSUPERVISED)**
- **RELIABLE AND EFFICIENT**

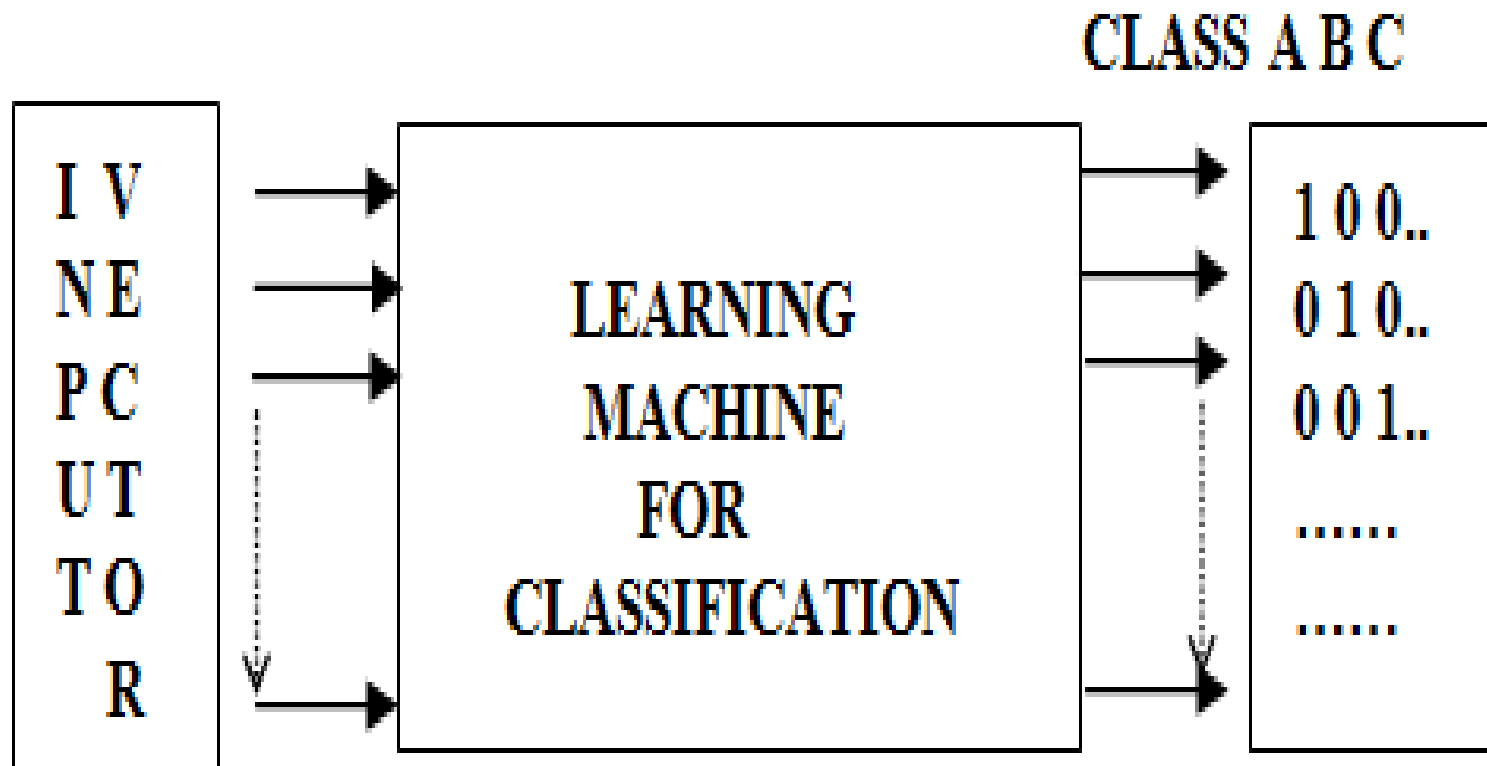
DISADVANTAGES OF MACHINE LEARNING

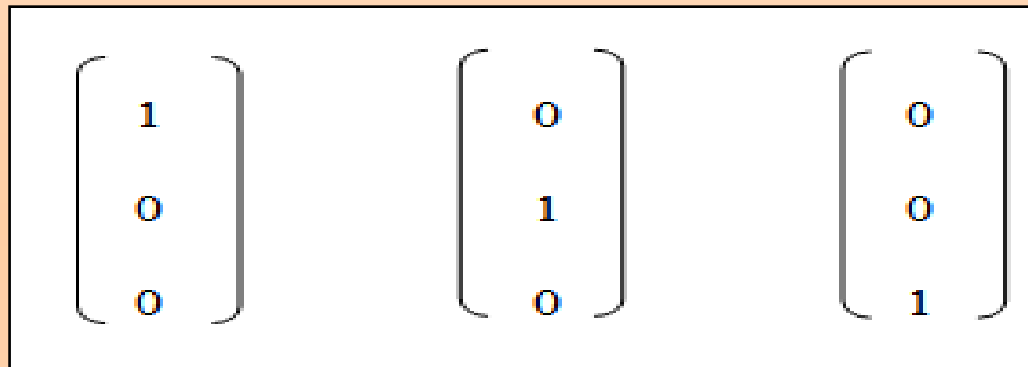
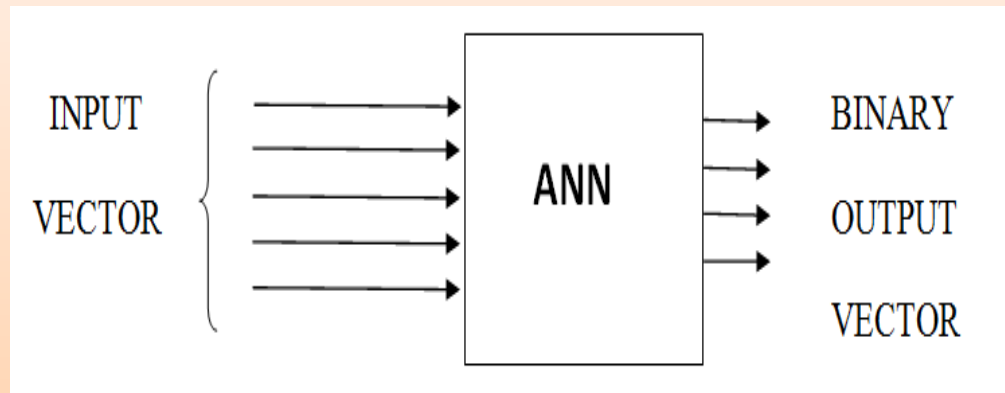
- **LACK OF EXPLAINABILITY**
- **HIGH CHANCES OF ERROR**
- **DATA BIAS**
- **COMPLEXITY**
- **MAKING THE WRONG ASSUMPTIONS**
- **INACCURATE PREDICTIONS**

HEURISTIC APPROACH

In problem solving tasks related to data mining those approaches are called heuristic which do not have explicit procedures to solve the specific problem rather we have to employ implicit techniques which may be a combination of novel mathematical and other scientific concepts that may be a part of computational algorithms. It is observed that there is no direct method available which decides the configuration of ANN for specific problem and dataset like number of hidden layers, number of neurons in these hidden layers, transfer function employed and other parameters which best fits the specific problem. Further cross-validation approach is based on partitioning of data in to multiple folds but there is no explicit reason available to decide number of folds for specific problem. In the experiments performed in this thesis we use hit and trial methods induced from the literature available from the various ANN resources.

DATA CLASSIFICATION



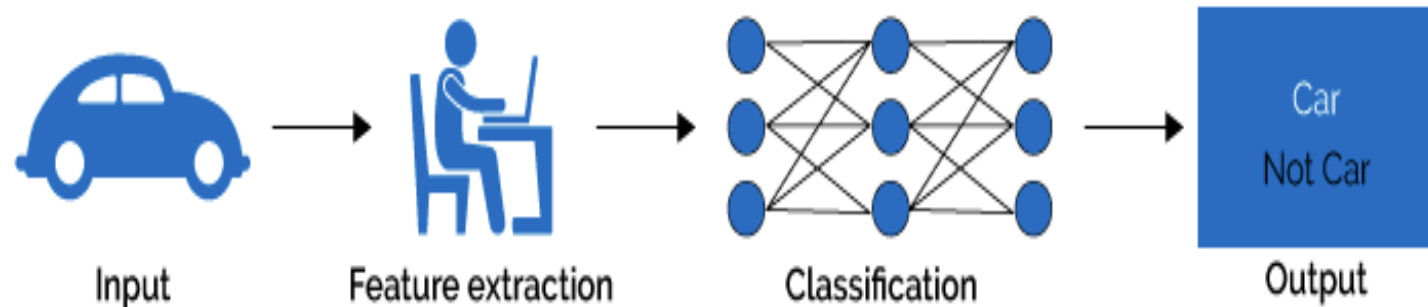


Output Vector	Interpretation
0.4	0
0.6	1
0.3	0

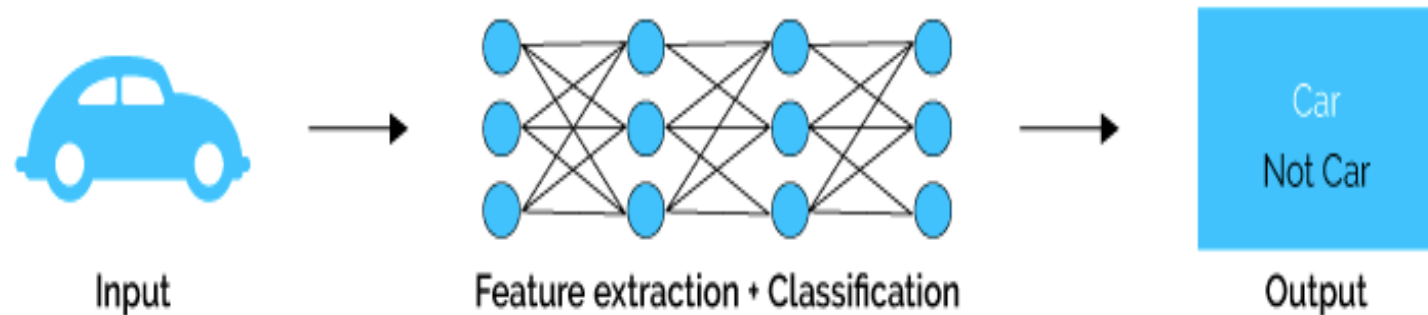
Output Vector	Interpretation
0.4	0
0.4	0
0.7	1

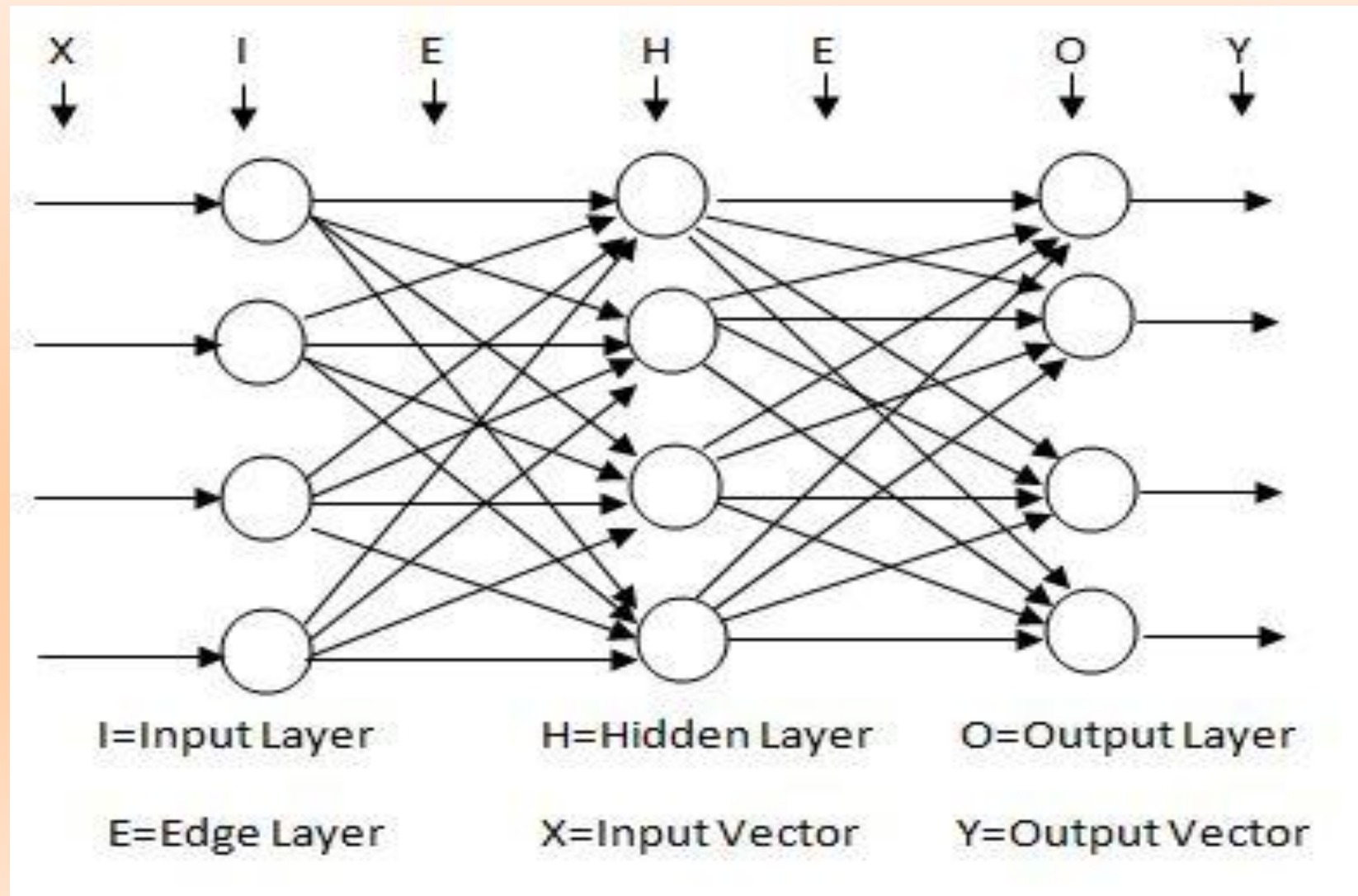
Output Vector	Interpretation
0.9	1
0.4	0
0.1	0

Machine Learning



Deep Learning





ARCHITECTURE OF ARTIFICIAL NEURAL NETWORK

THANK YOU